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ORIGINAL ARTICLE

Short-term traffic flow prediction: An ensemble machine learning approach

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Abstract The inconvenience of travel, air pollution and consequent economic losses caused by traffic congestion have seriously restricted the healthy and sustainable development of cities in China. In this context, as the main component of current and future urban traffic management measures, intelligent transportation system is an important means to improve the traffic efficiency of road network and alleviate urban traffic congestion. Traffic flow prediction plays an important role in this connection. In this paper, an ensemble short-term traffic flow prediction method based on optimized variational mode decomposition (OVMD) and combined long short-term memory network (LSTM) is proposed. The method consists of three main components: 1. Use the improved bat algorithm to optimize the parameters of VMD to achieve better decomposition effect; 2. Use the optimized variational mode decomposition algorithm (OVMD) to decompose the unstable original traffic flow time series data into relatively stable multiple Intrinsic Mode Functions (IMFs); 3. The optimized L-BILSTM model is established by combining the basic long short-term memory network with the bidirectional long short-term memory network. It can better extract information from traffic flow data and improve the accuracy of prediction results.

In the empirical study part, the traffic flow data of Changsha City is used to verify the prediction model proposed in this paper. The influence of the application of the variational mode decomposition algorithm to the training set data and the overall data on the final prediction results is also compared and analysed.

1. Introduction

In recent years, the rapid growth of vehicles has led to a large number of traffic problems, seriously affecting people's daily travel [1]. As a pivotal application of ITS, short-term traffic flow prediction plays a vital role in dynamic route planning [2], signal optimization [3], and real-time traffic management [4]. Thus, it constantly becomes one of the most crucial topics in traffic/transportation engineering. Traffic flow prediction is a typical regression problem of traffic network time series [5]. Based on the traffic data obtained by various detection equipment, combined with historical traffic data and other influencing factors, the prediction method and model are established. Accurate short-term traffic flow prediction is the basis and prerequisite for the operation of intelligent transportation sys-

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tems. Specifically, the current challenges encountered in short-term traffic flow prediction are mainly as follows.

(1) Although the combined model is more accurate in traffic flow prediction, its parameters are correspondingly more, and more computational effort must be spent to adjust the optimization model. Most of the existing combination models simply and roughly input the traffic flow data into the model for prediction, but there are few data integration studies.
(2) At present, most time series data processing methods do not consider the impact of the parameters of the method itself on the data, and most of them are determined by empirical or random methods.
(3) At present, most combination prediction models take into account the temporal and spatial attributes of the data itself, but do not consider the impact of the data on the model after dividing the training set and the test set. For example, when time series analysis is used, it is basically simple to decompose or stabilize all data, and then input it into the model. However, it ignores the negative impact of blind stacking algorithms and models, or that rough stacking models can indeed improve the prediction performance, but reasonable embedding can improve the rationality and efficiency of the model.

In order to solve these problems, this paper proposes a short-term traffic flow prediction method based on optimized variational mode decomposition algorithm OVMD and combined LSTM. In the prediction method proposed in this paper, the optimized VMD algorithm is used to decompose the input data, and then the optimized LSTM model is used to predict the decomposed traffic flow data. Specifically, OVMD decomposes the original non-stationary traffic flow data into relatively stable mode components and then inputs them into the prediction model. In general, the contributions of this paper are as follows.

(1) We propose an improved Bat algorithm, in which linear decreasing inertia weight is introduced. Compared with ordinary bat algorithm, this method can improve the problem of premature stagnation in the search process.
(2) Based on the improved bat algorithm, the OVMD algorithm is designed. OVMD algorithm is used to decompose the original traffic flow data. In this way, we remove the noise of the original data and obtain relatively stable time series data.
(3) We propose an optimized L-BILSTM model. This model effectively combines LSTM and BILSTM. The bidirectional LSTM makes up for the long-distance dependence of LSTM and makes full use of the forward and backward information of historical traffic data.
(4) It is verified on the data of Changsha. And the influence of OVMD applied to the training set data and the overall data on the final prediction result is compared and analyzed. The results show that the proposed model expresses a superior performance than the advanced baselines in terms of accuracy and efficiency. Additionally, the ablation experiment results also demonstrate the effectiveness of each component.

The rest of this paper is organized as follows. Section 2 reviews the existing research on short-term traffic flow prediction. We describe in detail the proposed short-term traffic flow prediction method based on the optimized variational mode decomposition algorithm OVMD and the combined LSTM in Section 3. In Section 4, we provide an extensive comparison of the short-term traffic flow prediction methods in this paper. Finally, we conclude this study and point out potential directions for future work in Section 5.

2. Related work

This section briefly introduces some related works, which can be grouped into three main types of research results in the field of short-term traffic flow prediction at this stage based on different prediction methods.

(1) Time series model based on statistics: In the early stage of research, scholars mostly used mathematical statistics to predict short-term traffic flow [6,7]. The collected traffic flow data is regarded as time series, and the time series model is used for prediction. For example, Thomas et al. established the univariate and multivariate ARIMA models based on the collected real-time traffic flow data, and compared them, and finally found that the performance of the multivariate model was better [8]. Comert and Bezuglov built a multi-step adaptive prediction model based on ARIMA, and the final prediction showed that the constructed model was superior to the previous model [9]. With the gradual deepening of the research on short-term traffic flow prediction, scholars began to notice the spatiotemporal characteristics of traffic flow data itself [10], but most of these statistical models based on premise assumptions are too idealistic, and can not deal well with complex models in real traffic flow [11,12].
(2) Traffic flow prediction models based on artificial intelligence: With the rise of artificial intelligence, many artificial intelligence models are increasingly used in the transportation industry. Chan et al. improved the stability and reliability of the prediction model by optimizing the particle swarm algorithm [13]. Hong et al. used a differential evolutionary algorithm to optimize the parameters in a neural network model, and the resulting traffic flow prediction model outperformed ordinary artificial neural networks in terms of performance [14]. Xu et al. used the continuous ant colony optimization algorithm (SVRCACO) to optimize support vector regression (SVR), and the resulting model performed well in short-term traffic flow prediction [15]. It can be seen that neural network is deeply loved by scholars in traffic flow prediction [16–20]. The neural network has good effects on the prediction of nonlinear data and the mining of complex patterns of traffic flow data [21]. MA et al. applied the long short-term memory model to the field of traffic flow prediction, which overcomes the problem of back propagation error attenuation of the memory module [22]. Tian et al. confirmed the advantages of recurrent neural network in prediction through example verification and comparison of several models [23]. Song et al. also used particle swarm optimization as a tool to optimize the chaotic time series of BP neural network[24]. With the rise of big data and deep learning, the figure of deep learning in traffic flow prediction is also appearing more and more, which can capture features from massive historical traffic data [25]. Lv et al. used stacked autoen-

coders to extract traffic flow features [26]. In [27,28], hybrid long short-term memory networks started to be applied to the simulation and prediction of traffic flows. Zhao et al. proposed a new LSTM network in which the two-dimensional direct representation of spatiotemporal correlations [29]. To obtain spatiotemporal traffic information in the transport network topology, convolutional LSTMs [30] and graph convolutional gated recurrent units [31] are used to determine temporal relationships. Tian et al. proposed a long short-term memory model for the problems of missing traffic flow data and irregular sampling [32]. However, AI models still suffer from shortcomings such as parameter selection and overfitting problems [33–35].

(3) Hybrid model: A single model has its limitations, such as difficulty in capturing dynamic fluctuations and some characteristics of traffic flow. For example, temporal characteristics and spatial characteristics [36,37]. On the basis of summarizing the predecessors, later scholars began to combine and improve the single prediction model to compensate for these limitations [38]. For example, Luo et al. input traffic flow data into a KNN model to obtain the output results and then input them into a long short-term memory network for secondary training [39]. Tan et al. combined ARIMA with an artificial neural network to form a combined model that could exploit the time-series nature of traffic flow data [40]. Chen et al. combined ANN and fuzzy inference system and further optimized the hybrid model using particle swarm optimization (PSO) algorithm based on this [41]. It can be seen that PSO is widely used in optimizing traffic prediction models [42]. Although the hybrid model solves some problems that cannot be solved by a single model, its feature mining of data itself is still limited. So scholars further combine data feature analysis with prediction model. Dai et al. combined spatiotemporal analysis with gated recurrent units, which greatly improved the accuracy of short-term traffic flow prediction [43]. Tipping proposed a sparse Bayesian learning model Relevance Vector Machine (RVM) [44]. Chan et al. used scale analysis to split the historical traffic flow data and finally used a combined model to make predictions [45].

Although hybrid models are more accurate in traffic flow prediction, they have more parameters and must spend more computing power to adjust and optimize the model [46]. In addition, most of the existing hybrid models simply input the traffic flow data into the model for prediction, and there is little research on the integration of the data itself.

3. Methodology

3.1. Variational mode decomposition (VMD)

Variational mode decomposition (VMD) is a method of signal smoothing. It can smooth the original nonlinear historical traffic flow data and decompose it into multiple pieces of relatively stable data. This is very important in traffic flow prediction. Because most of the prediction models are not very good at predicting non-stationary data, on the contrary, if the input model is smooth and stable data, the prediction effect of the model can be greatly improved.

The overall framework of VMD is a variational problem, which decomposes non-stationary signals into several intrinsic mode fractionss with limited bandwidth.

(1) The establishment of the variational problem

The one-sided spectrum is obtained by first computing the analytical signal of each mode function with the Hilbert transform; The mode spectrum is then converted to baseband, and the corresponding estimated center frequency is obtained by using exponential adjustment.

$$\begin{cases} \min_{(u_k),\left(\omega_{k)\right.}} \left\{ \sum_k \| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k^t} \|_2^2 \right\} \\ s.t. \sum_k u_k = f \end{cases} \tag{1}$$

where: u_k is the mode fractions obtained by signal decomposition, $\hat{u}_k$ is the center frequency of mode fractions, K is the number of iterations (k = 1,2,3,...),$\delta(t)$ is the Dirac distribution, and * represents the convolution operation. ω_k is the center frequency, f is the original signal.

(2) Solving the variational problem.

In order to solve the variational problem of Eq. (1), Lagrange's formula is used for the solution:

$$\begin{aligned} L(\{u_k\},\{\omega_k\},\lambda) := \partial \sum_k \| \partial_t \left[\left(\delta(t) + \tfrac{j}{\pi t} \right) u_k(t) \right] e^{-j\omega_k^t} \|_2^2 + \\ \| f(t) - \sum_k u_k(t) \|_2^2 + \left\langle \lambda(t), f(t) - \sum_k u_k(t) \right\rangle \end{aligned} \tag{2}$$

∂ and are both the penalty factor, λ is used to maintain the stringency of the constraints. The optimal solution of the constrained variational model of Eq. (1) is found by iterating the suboptimization sequence. This leads to:

$$\begin{aligned} \hat{u}_k^{n+1}(\omega) &= \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{u}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\partial(\omega - \omega_k)^2} \\ \omega_k^{n+1} &= \frac{\int_0^\infty \omega |\hat{u}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_k(\omega)|^2 d\omega} \end{aligned} \tag{3}$$

The implementation process of the VMD method is as follows:

I. Initialize parameters $\{\hat{u}_k^1\}$, $\{\omega_k^1\}$, $\{\hat{\lambda}^1\}$, and set the number of iterations n to 1.

II. According to formulas (3) and (4), update u_k,ω_k.

III. Update the Lagrange penalty operator according to formula (5):

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau \left(\hat{u}(\omega) - \sum_k \hat{u}_k^{n+1}(\omega) \right) \tag{4}$$

where:τ is the update parameter.

According to formula (6), judge whether the iterative stop condition is met, if so, stop the iteration and output the result; otherwise, the iteration number n is increased by 1, and return to step II.

$$\sum_k \| \hat{u}_k^{n+1} - \hat{u}_k^n \| \| \| \hat{u}_k^n \|_2^2 < \varepsilon \tag{5}$$

Obtain the corresponding mode subsequence according to the given mode number.

3.2. The improved bat algorithm

3.2.1. The bat algorithm

The bat algorithm (BA) is a swarm intelligence optimization algorithm [47]. The BA algorithm measures the fitness value of the objective function of the solution problem as the superiority or inferiority of the position in which the individual bat is located. The optimal solution is analogous to the bat's prey, and the change of pulse loudness and frequency indicates the proximity to the optimal solution to some extent.

In the BA algorithm, the search frequency, position and velocity are updated as follows:

$$p_i = p_{min} + (p_{max} - p_{min})\partial \tag{6}$$

$$V_i^{t+1} = V_i^t + \left(S_i^t - S^*\right)p_i \tag{7}$$

$$S_i^{t+1} = S_i^t + V_i^{t+1} \tag{8}$$

where:p_i、p_{max} and p_{min} denote the acoustic frequency emitted by the ith bat at the current moment and the maximum and minimum values of the acoustic frequency, respectively; a random number of$\partial \in [0,1]$; and the current optimal position.

$$X_{new} = X_{old} + \beta A^t \tag{9}$$

where: X_{old} is a solution chosen from the current optimal solution; A^t is the average of all bat loudnesses at time t; β is a random number of [-1,1]. The loudness Ai and the rate of bat pulses are calculated by the following equation:

$$A_i^{t+1} = \gamma A_i^t \tag{10}$$

$$l_i^{t+1} = l_i^0(1 - e^{-\varepsilon t}) \tag{11}$$

where: γ and ε are constants and $0 < \gamma < 1$, $\varepsilon > 0$. The continuous iterative updates of the loudness Ai and rate l_i indicate the proximity to the optimal solution.

The bat algorithm is able to achieve fast convergence of the algorithm by switching from global to local optimization in the early stages of the algorithm operation, which can also lead to a premature stagnation phase of the algorithm. To prevent the algorithm from stalling prematurely, this paper incorporates linearly decreasing inertia weights in the velocity update:

$$v_i^{t+1} = w(n)v_i^t + \left(s_i^t - s^*\right)p_i \tag{12}$$

$$w(n) = w_{\max}(w_{max} - w_{min})(T_{max} - n)/T_{max} \tag{13}$$

where:$w_{\max} = 0.89$, $w_{min} = 0.41$ [48]. n is the current number of iterations; T_{max} is the maximum number of iterations. $w_{\max}$ is the inertia weight at the beginning of the iteration, the larger its value, the more powerful the global search capability, and the inertia weight decreases linearly as the number of iterations increases. w_{min} is the inertia weight at the end of the iteration, which provides better local search As the inertia weight decreases linearly with the number of iterations, w_{min} as the inertia weight in the late iteration can provide better local search capability and avoid premature stagnation of the algorithm.

3.2.2. Fitness function

Before VMD decomposition, the number of components K and penalty factor C need to be preset. Different values of K and C have different decomposition results. When the values of K and C are too large, false components will be generated and the component bandwidth will be too small. When the values of K and C are too small, it will lead to mode aliasing and excessive component bandwidth.

The parameter optimization of VMD using the bat algorithm requires the selection of an adaptive function. The adaptive function in this paper uses the very small envelope entropy $f(g)$:

$$f(g) = -\sum_{j=1}^{N} g_i \lg(g_i) \tag{14}$$

$$g_i = \frac{\theta(j)}{\sum_{j=1}^{N}\theta(j)} \tag{15}$$

$$\theta(j) = \sqrt{h^2(j) + \widehat{h}^2(j)} \tag{16}$$

where:g_i is the sequence of probability distributions of $\theta(j)$; $\theta(j)$ is the envelope signal obtained after Hilbert demodulation of signal $h(j)$.

The process of using the improved BA algorithm to optimize VMD is shown in Fig. 1.

The specific pseudocode is listed in Algorithm 1.

Algorithm 1 Framework of VMD optimized by BA

Fitness functions:$f(g) = -\sum_{j=1}^{N} g_i \lg(g_i)$
Input: The population size N, repository size R and maximum number of iterations T
Output: The variates K, C, and the objective function values $f(g)$
1: Initialize: The random population i including two variates K ∈ (2,10), C∈ (1000, 3000)
2: while (t < T)
3: Generate new solutions
4: Update speed and position/solution [Eqs. (1) to (3)]
5: if (rand1 > ri)
6: Select a solution
7: Randomly generate a local solution in the vicinity of the chosen solution
8: end if
9: Generate new solutions by random flight
10: if (rand2 < Ai & f(xi) < f(x*))
11: Accepting new solutions
12: Increase pulse firing rate
13: Reduce loudness
14: end if
15: Find the current best solution
16:end while

In general, the benefits and drawbacks of the above methods are shown in Table 1.

3.3. Long Short-Term memory network (LSTM)

The disadvantage of recurrent neural networks (RNNs) is that they have a limited range of contextual information available. Therefore, with the increasing range of recursion, the internal input will slowly lose its impact on the network output, and ultimately affect the performance of the entire network. To address this drawback, LSTM was born. Which improves the ability to obtain upstream and downstream information of the data but is not a perfect structure because LSTM only

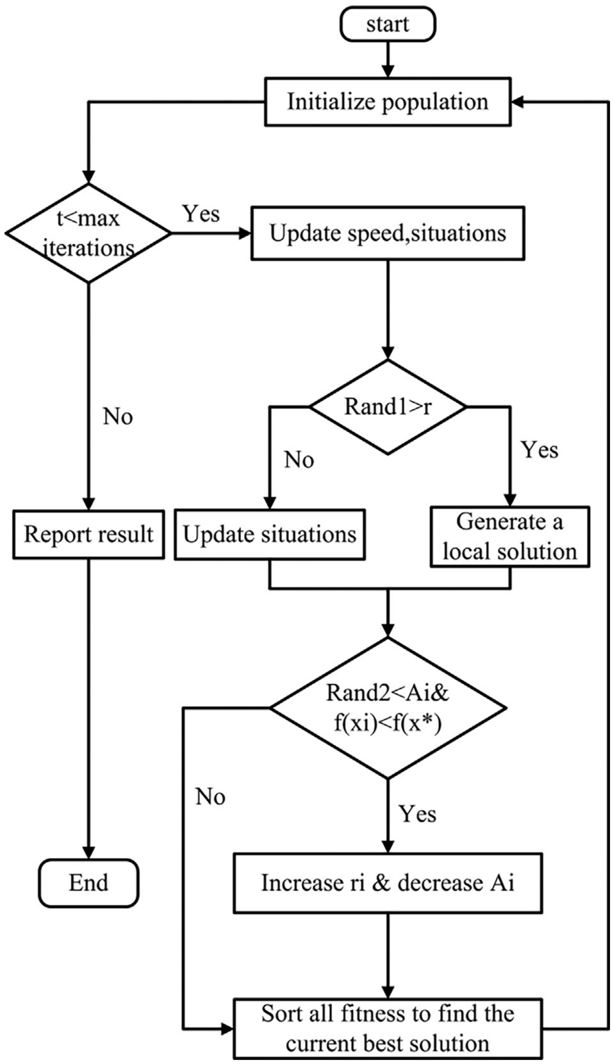

Fig. 1 VMD optimization process.

uses the forward information in the time series and has poor access to the backward information. In practical applications, both the forward and backward data contain important information that needs to be acquired. On this basis, a bidirectional LSTM neural network is proposed. It can mine the context information in the data at the same time, so that the network can learn faster and more fully.

Table 1 Comparison between improved methods and existing methods.

	Benefits	Drawbacks
Bat algorithm	efficient	Easy to fall into premature stagnation
Improved bat algorithm	Improve the problem of premature stagnation in the search process	Slightly high complexity
VMD	Fast	Unable to determine appropriate parameters
OVMD	Convenient to determine appropriate parameters	Slightly high complexity

As shown in Fig. 2, LSTM is a chain structure of repeating neural network modules. X is the input, h is the weight of the neuron, and tanh is the activation function.

The network neuron structure of LSTM is shown in Fig. 3. In each memory cell (A in the Fig. 2) includes the forget gate, input gate and output gate.

Forget gate: As shown in Fig. 4, the role of forget gate in LSTM is mainly used to forget the previous state, which is used to get what needs to be removed from the previous state by the sigmoid activation function.

$$f\_t = \sigma\left(W_f \cdot [h\_t - 1, x\_t] + b_f\right) \tag{17}$$

Input gate: As shown in Fig. 5. The function of the input gate is to process the input of the current sequence position and update the cell status. This process is divided into two parts. One part is to use the input gate containing sigmoid layer to determine which new information should be added to the cell state. After determining which new information to add, it is necessary to convert the new information into a form that can be added to the cell state. The other part is to use the tanh function to generate a new candidate vector. Input threshold layer i_t Determine the unit status to be updated.

$$i_t = \sigma(W_i \cdot [h\_t - 1, x\_t] + b_i) \tag{18}$$

$$C_t' = \tanh(W_C \cdot [h\_t - 1, x\_t] + b_C) \tag{19}$$

As shown in the Fig. 6, where f_t × C_t − 1 represents the information to be deleted, and i_t × C_t′ represents the newly added information.

$$C_t = f\_t * C\_t - 1 + i\_t * C\_t' \tag{20}$$

Output gate: As shown in the Fig. 7**.** It is the same as the update of the input gate. The output gate also needs to use the sigmoid activation function to determine which part of the content needs to be output, and then use the tanh activation function to process the content of cell state (because each value of Ct calculated above is not in the range of tanh − 1 ~ 1, which needs to be adjusted).

$$O_t = \sigma(W_O \cdot [h\_t - 1, x\_t] + b_O) \tag{21}$$

$$ht = O\_t * \tanh(C\_t) \tag{22}$$

3.4. Bidirectional LSTM

The bidirectional LSTM is a combination of forward LSTM and reverse LSTM. Bidirectional LSTM can not only predict the output of the next moment based on the time sequence information of the previous moment, but also predict the output of the next moment by combining the future state and historical information.

The bidirectional LSTM replaces the normal RNN units in the BiRNN with LSTM units. Its structure is shown in Fig. 8.

The data will undergo a round of forward computation after entering the Forward layer and a round of reverse computation after entering the Backward layer, and the results of the two layers will be processed and then output.

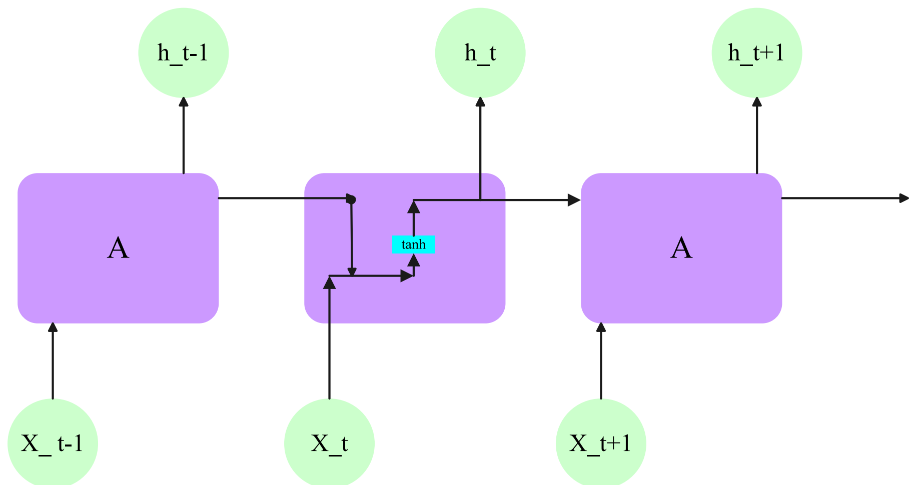

Fig. 2 LSTM network structure.

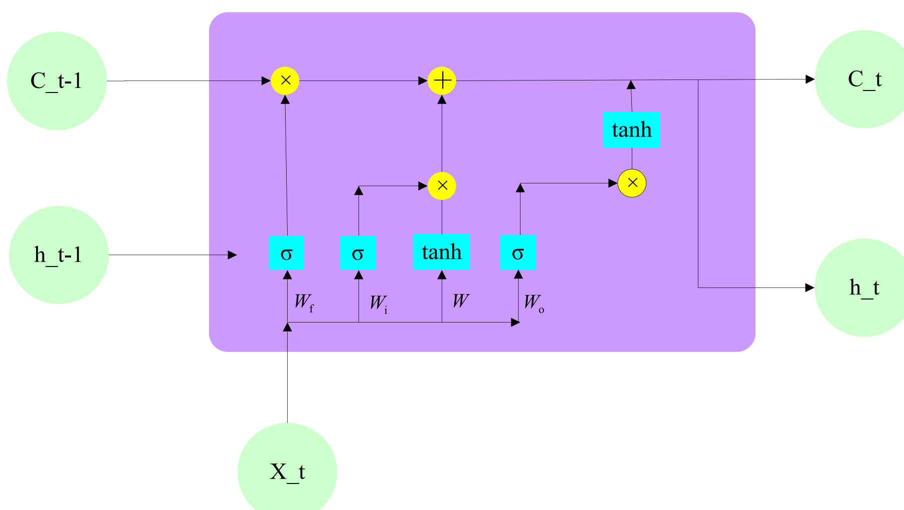

Fig. 3 Neuron structure of LSTM neural network.

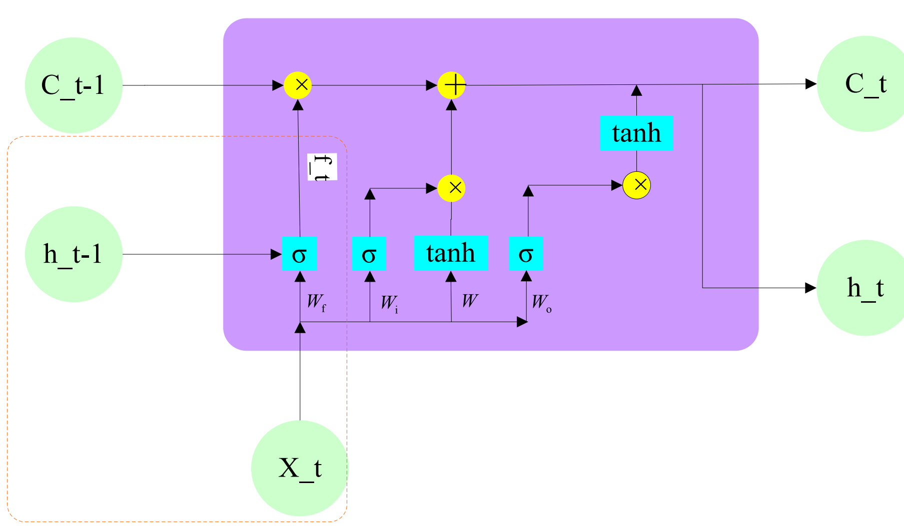

Fig. 4 Forget gate.

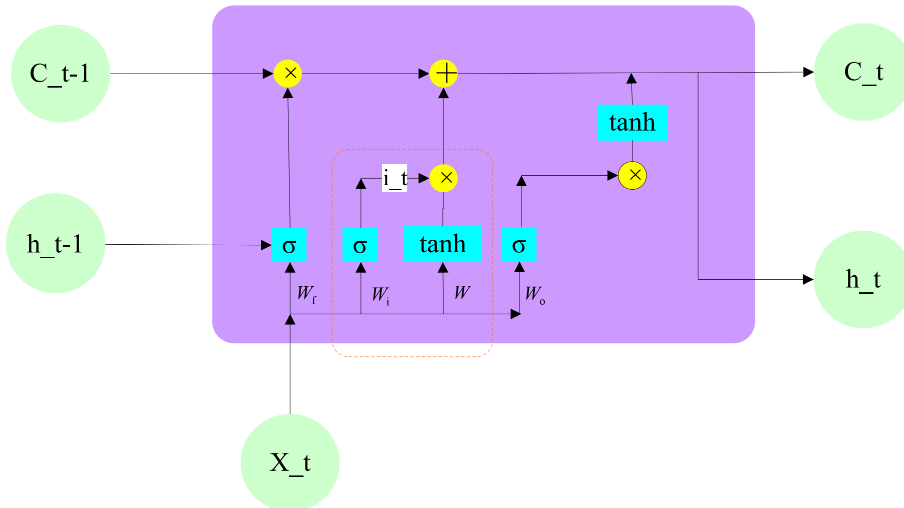

Fig. 5 Input gate.

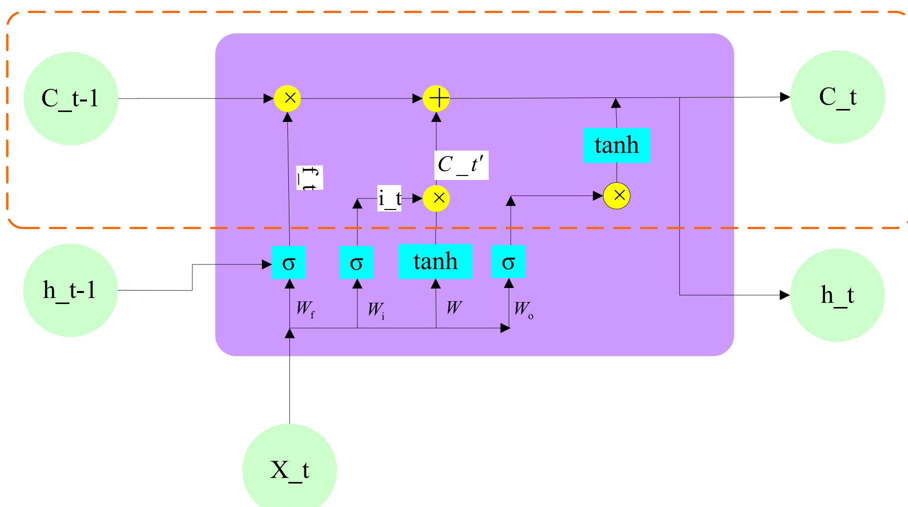

Fig. 6 Update process.

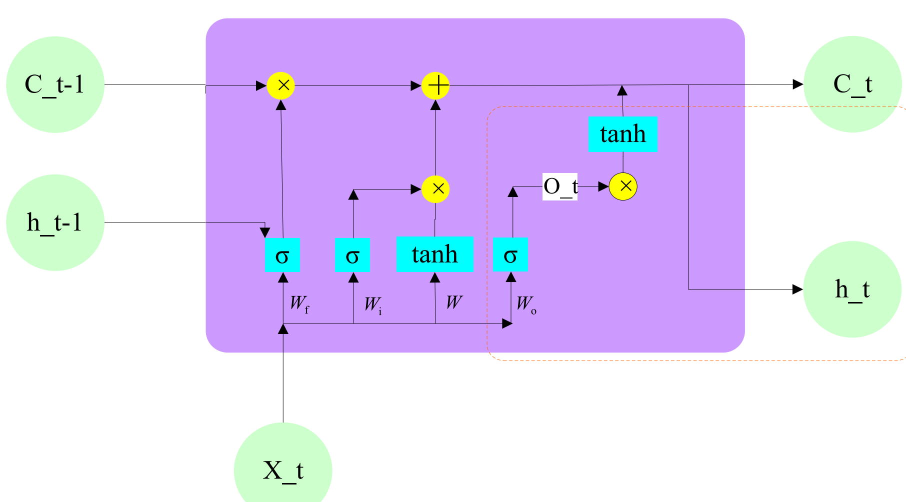

Fig. 7 Output gate.

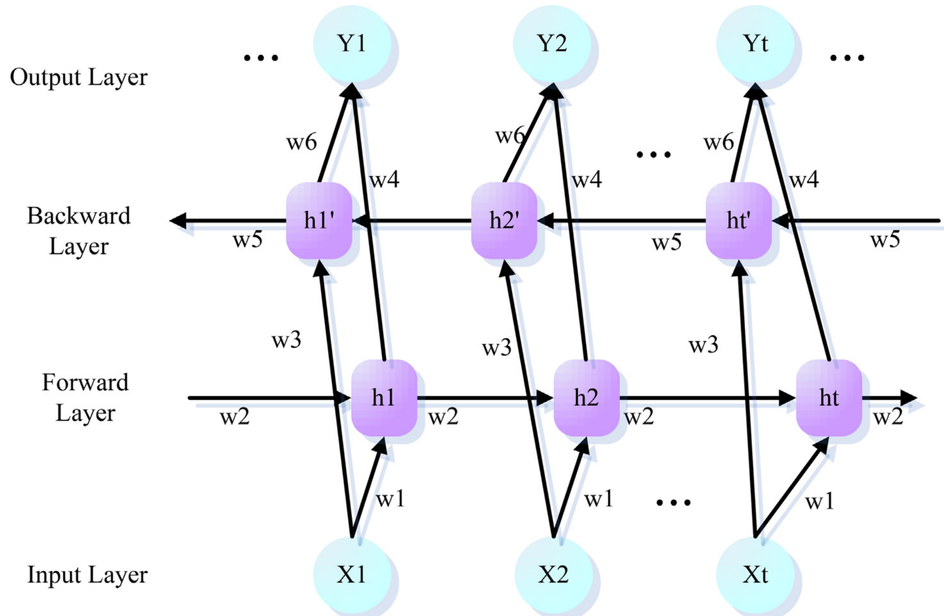

Fig. 8 Bidirectional LSTM network structure.

3.5. Model construction

In this paper, we use the optimized LSTM model shown in Fig. 9, which consists of a two-layer LSTM and a two-layer bidirectional LSTM, where Dense is the fully connected layer. The optimizer in this model is "Nadam" and the activation function is "tanh".

The prediction process is shown in Fig. 10. First, the OVMD algorithm is used to decompose the non-stationary historical traffic flow data into smooth Intrinsic Mode Functions (IMF); Then input the decomposed IMF into the optimized LSTM model for learning and training; Output and overlay the predicted value of each IMF to obtain the final traffic flow prediction result.

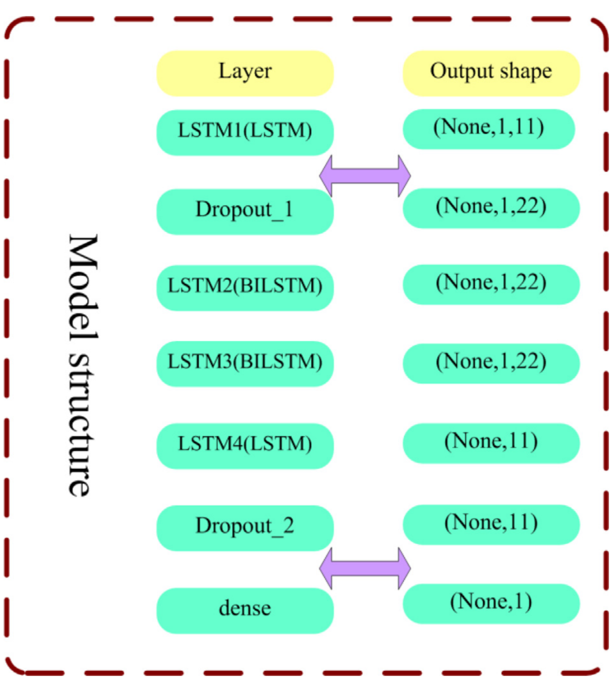

Fig. 9 Model construction.

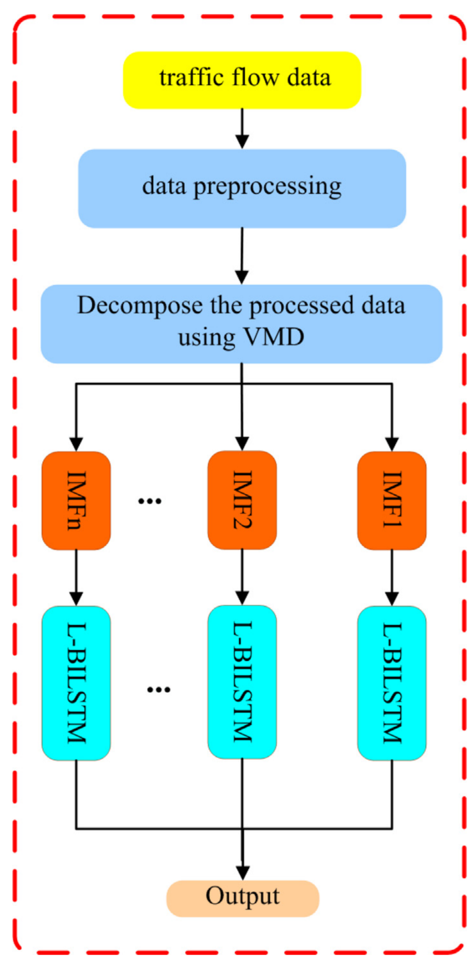

Fig. 10 Prediction process.

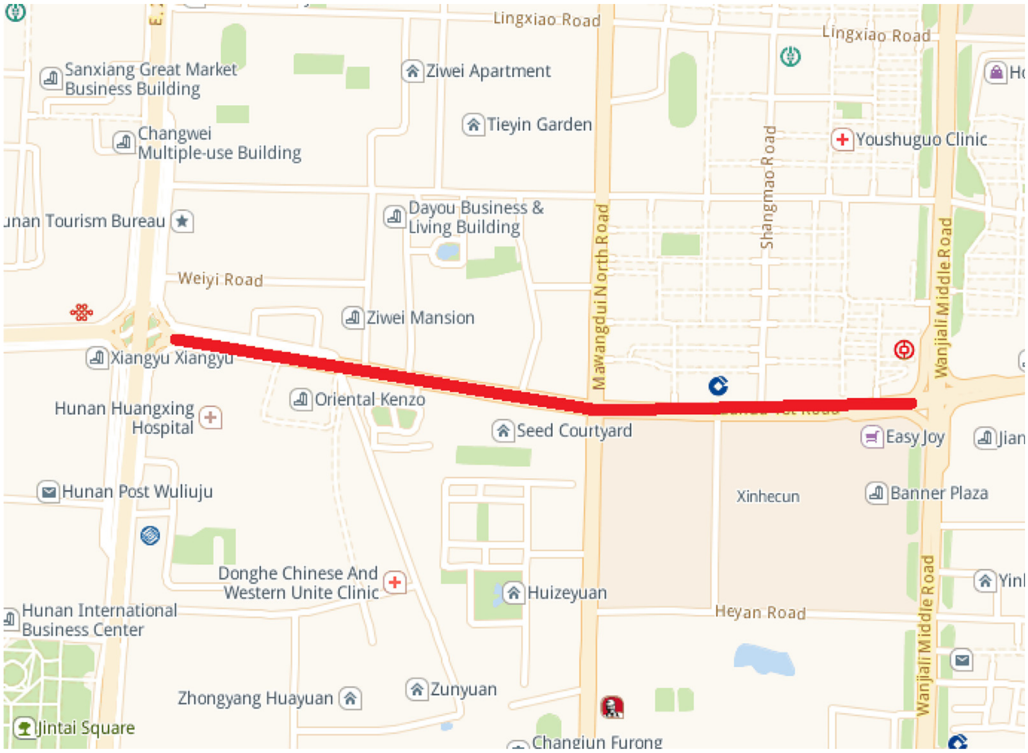

Fig. 11 Road sections selected from experimental data.

4. Empirical study

4.1. Data sources

The data selected in this paper is the traffic flow data of Yuanda Road, Furong District, Changsha City, from September to October in 2013, excluding weekend traffic data, for a total of 40 days. Specific road sections are shown in Fig. 11. The data comes from real-time collection of toroid in each lane at each intersection. The traffic data is output by the SCATS Traffic Reporter system, with 5 min as the collection period, and each detector collects 288 traffic data every day.

4.2. Evaluation function

In this paper, MAE, RMSE are selected as the accuracy evaluation indicators for comparing these prediction algorithms.

MAE is the mean absolute error, which is calculated as follows:

$$MAE = \frac{1}{n}\sum\nolimits_{i=1}^{n} |z * -z| \tag{23}$$

RMSE is the root mean square error, which is calculated as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum\nolimits_{i=1}^{n} (z * -z)^2} \tag{24}$$

In the above equation, z* is the predicted traffic flow and z is the actual traffic flow.

4.3. VMD decomposition

The optimized combination of VMD parameters using the improved bat algorithm is shown in Table 2.

It can be seen that the minimal envelope entropy has the smallest value when the number of iterations is 60. Therefore, this paper chooses K = 6. The default value of the penalty parameter C is 2000, and the value of $\hat{o}$ is 0.3 to ensure that the data is not distorted. The original load sequence and its subsequence effect obtained by VMD decomposition are shown in the Fig. 12.

4.4. Comparative analysis

This experiment discusses the influence of the optimized variational mode decomposition algorithm (OVMD) on the accu-

Table 2 Optimization of different parameter combinations.

Iteration	K	C	Minimal envelope entropy
20	1736	8	3.3318
30	2196	6	3.3229
40	2250	6	3.3323
50	1927	7	3.3286
60	2079	6	3.3217
70	1856	7	3.3225
80	2317	6	3.3312
90	2647	6	3.3362
100	2532	6	3.3351

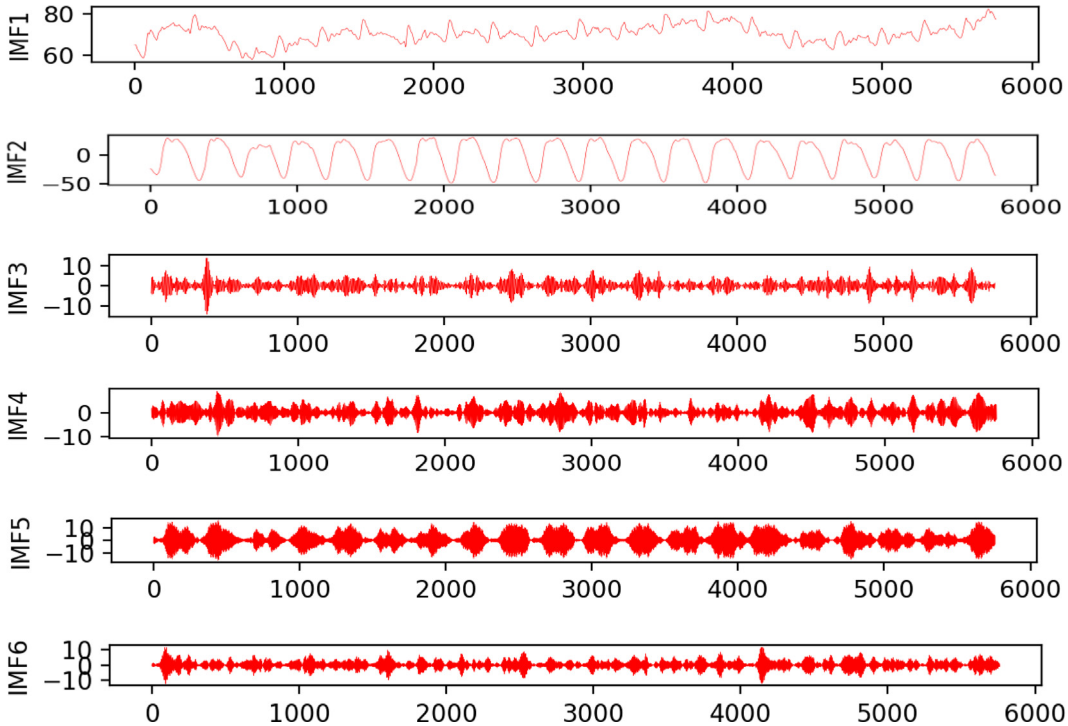

Fig. 12 VMD decomposition effect.

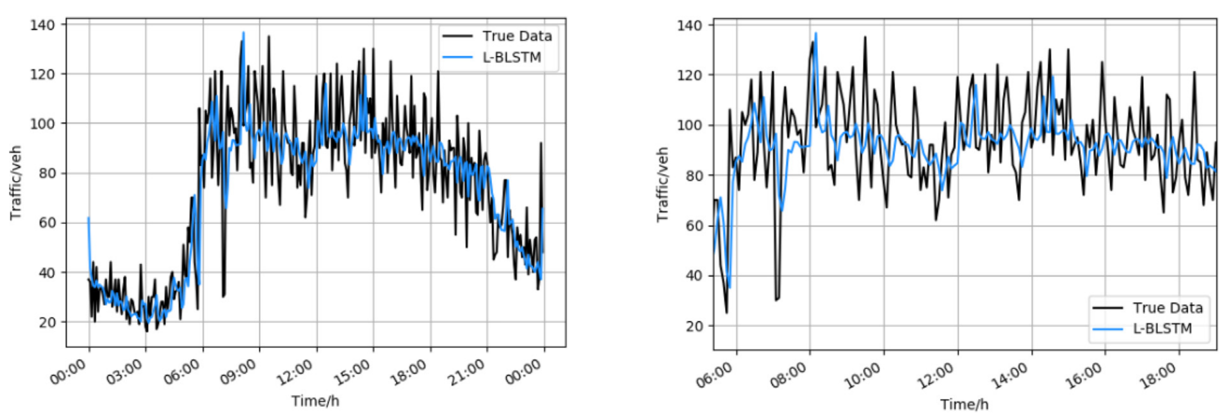

Fig.13 L-BILSTM model prediction results.

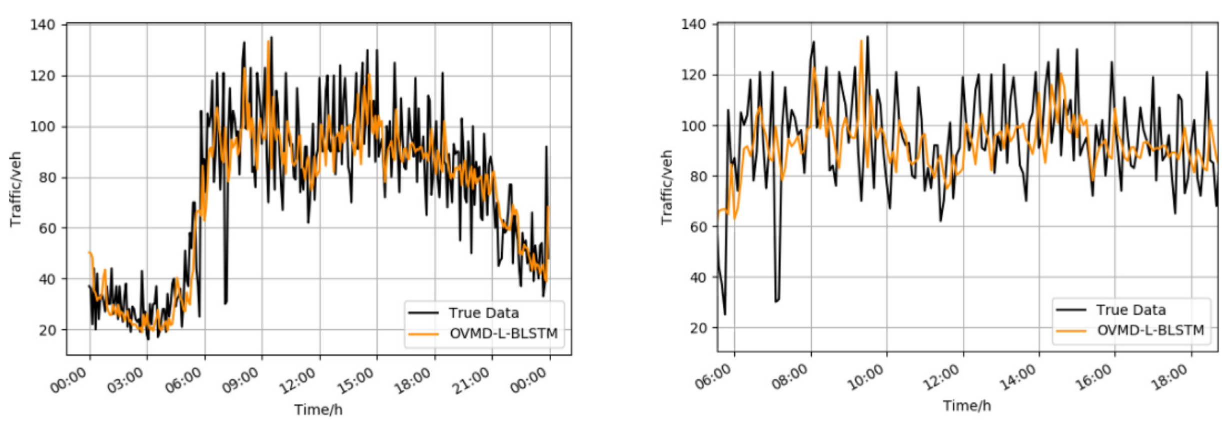

Fig. 14 OVMD-L-BILSTM model prediction results.

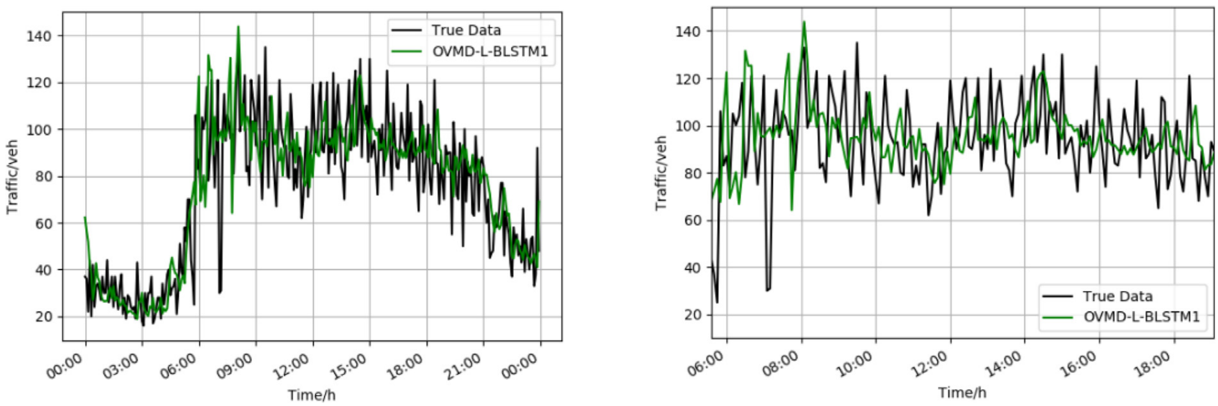

Fig. 15 OVMD-L-BILSTM1 model prediction results.

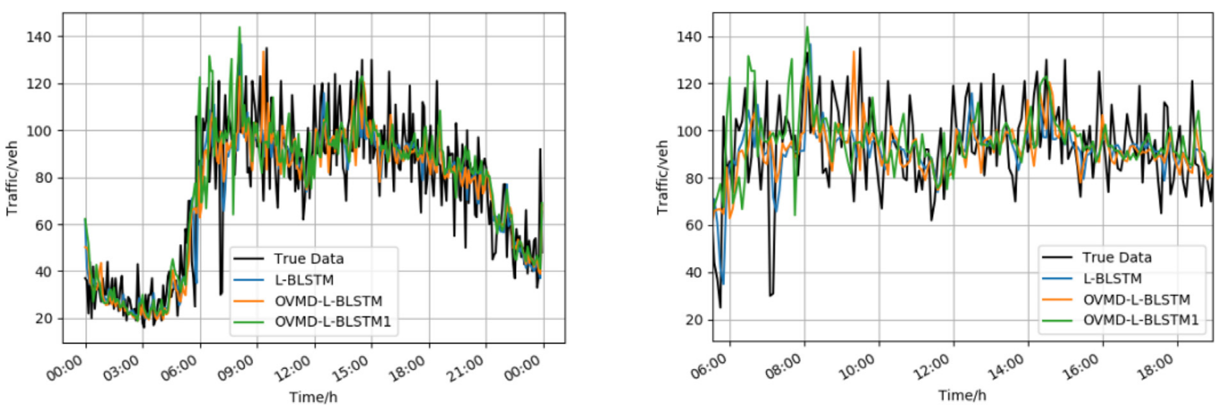

Fig. 16 Comparison of three groups of experiments.

racy of traffic flow prediction, and compares three groups of experiments. The first group (L-BILSTM): input the original traffic flow data directly into the L-BILSTM model for prediction. The second group (OVMD-L-BILSTM1): all traffic flow data are decomposed by variational mode and then input into L-BILSTM model for prediction. The third group (OVMD-L-BILSTM): only decompose the training data into L-BILSTM model to predict the value of each IMF fractions, and finally integrate the prediction test set.

Figs. 13, 14, 15 and 16 are the results of three groups of experiments.

It can be seen from Table 3 that the third group performs best in the prediction of the five-minute time step. That is, only the training data is decomposed, and the improved long short-term memory network model is included to predict the value of each IMF component, and finally the prediction test set is integrated. Its MAE value of 12.95 is lower than the second group's 13.52 and the first group's 14.88. The RMSE value is also the best among the three models. The effect of the second group on variable-mode decomposition of all traffic flow data and then inputting the improved LSTM model for prediction is also slightly better than that of the first group using only the improved LSTM model. But the second group performed worse than the third group. These three sets of comparative tests show that OVMD plays a very important role in traffic flow prediction. If only the training set is decomposed into variational modes, the final prediction effect will be better.

In order to prove that the OVMD algorithm can improve the performance of the prediction model, this paper also designs several other groups of commonly used short-term traffic flow prediction models to compare the prediction results of the OVMD algorithm with the original model. And the comparative experiment of using OVMD decomposition only for training data and OVMD decomposition for all data. Among them: OVMD-SAE1, OVMD-GRU1, OVMD-L-BILSTM1 uses OVMD decomposition for all data and OVMD-SAE, OVMD-GRU, OVMD-L-BILSTM uses OVMD decomposition for training data. Fig. 17 **and** Fig. 18 are boxplot of the prediction results of each model.

As can be seen in Table 4, the performance of all models decreases as the time step increases. The prediction performance of each model at multiple time steps is more stable after the inclusion of OVMD decomposition in Fig. 19. Among these selected models again, the OVMD-L-BILSTM has the most outstanding prediction performance. While the SAE model at a time step of 20 min, the prediction model that uses OVMD decomposition for all data performs better than the prediction model that only uses OVMD decomposition for training data. And throughout all the models, the prediction model that uses OVMD decomposition for all data is more stable in the interval of 10 min, 15 min and 20 min. The prediction model using OVMD decomposition only for the training data has a more severe performance degradation after increasing the time step.

Table 3 Experimental results.

model	MAE	RMSE
L-BILSTM	14.88	21.33
OVMD-L-BILSTM1	13.52	18.99
OVMD-L-BILSTM	12.95	17.61

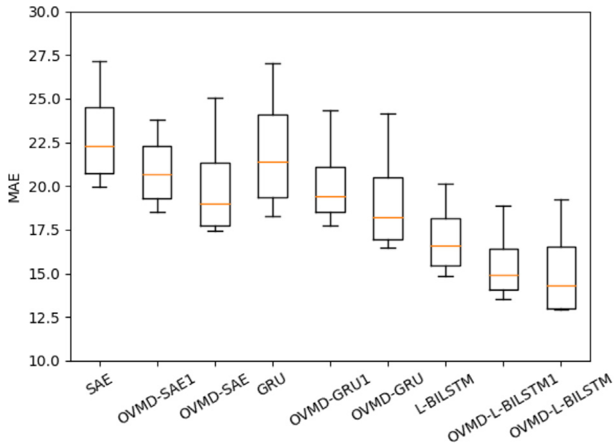

Fig. 17 The MAE boxplot of each model.

Therefore, it can be concluded that the performance of the OVMD-L-BILSTM model proposed in this paper is better in the case of short-term traffic flow prediction such as five minutes and ten minutes.

4.5. Ablation experiment

To clarify the effect of each component on the model, a comparative analysis is performed using the improved bat algorithm optimized OVMD, ordinary VMD and separate prediction models, OVMD-L-BILSTM, VMD-L-BILSTM and L-BILSTM. Only the training data is decomposed by OVMD. In the experiment, 5 min, 10 min, 15 min, 20 min, 25 min and 30 min correspond to six time steps. The specific comparison results are shown in Fig. 20.

As can be seen in Fig. 20, the prediction performance of OVMD-L-BILSTM is better relative to the other two models. And its comprehensive performance is also to be due to its variant model. The results show that the OVMD decomposition algorithm has a significant impact on the prediction results, and the optimized OVMD algorithm using the improved bat algorithm has a greater impact on the prediction model than the ordinary VMD algorithm. This illustrates the significance of the optimized OVMD algorithm and proves the effectiveness of the model.

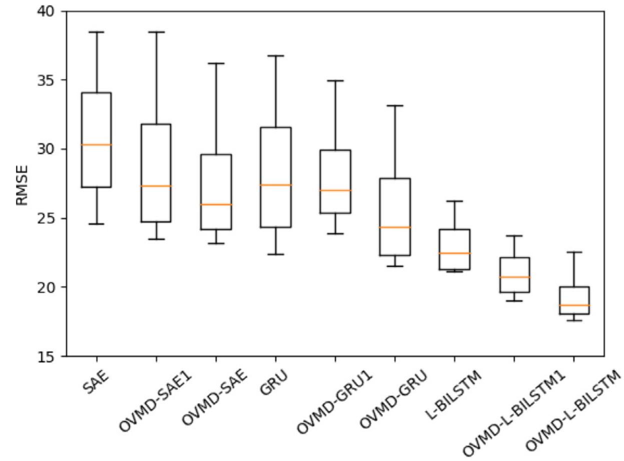

Fig. 18 The RMSE boxplot of each model.

5. Conclusion

In this paper, an ensemble machine learning method is proposed to predict short-term traffic flow. It is different from previous studies that directly optimize the prediction model. We propose an optimized variational mode decomposition algorithm (OVMD). First, we use the improved Bat algorithm to solve the optimal parameters of VMD, and then use OVMD to decompose the original traffic flow data. Then we establish an optimized L-BILSTM model based on the long short-term memory network and combined with the bidirectional long short-term memory network. Finally, we use the traffic flow data of Changsha City to verify the prediction model proposed in this paper. The influence of the application of the variational modal decomposition algorithm to the training set data and the overall data on the final prediction results is also compared and analyzed.

However, there are also shortcomings in this paper. We did not consider the impact of multivariate factors on the prediction model, such as weather, holidays and other factors. It also

Table 4 Prediction results of each model under four time steps.

	5 min		10 min		15 min		20 min	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
SAE	19.96	24.58	21.01	28.11	23.61	32.59	27.17	38.51
OVMD-SAE1	18.53	23.46	19.52	25.12	21.79	29.57	**23.82**	**33.92**
OVMD-SAE	**17.43**	**23.19**	**17.87**	**24.55**	**20.11**	**27.41**	25.03	36.21
GRU	18.26	22.36	19.68	25.02	23.15	29.86	27.01	36.77
OVMD-GRU1	17.72	23.88	18.76	25.89	20.03	28.21	24.31	34.95
OVMD-GRU	**16.46**	**21.5**	**17.1**	**22.56**	**19.29**	**26.17**	**24.18**	**33.1**
L-BILSTM	14.88	21.33	15.66	21.12	17.53	23.55	20.11	26.22
OVMD-L-BILSTM1	13.52	18.99	14.22	19.21	15.61	19.88	18.88	23.69
OVMD-L-BILSTM	**12.95**	**17.61**	**12.99**	**18.23**	**14.66**	**19.22**	**16.21**	**22.53**

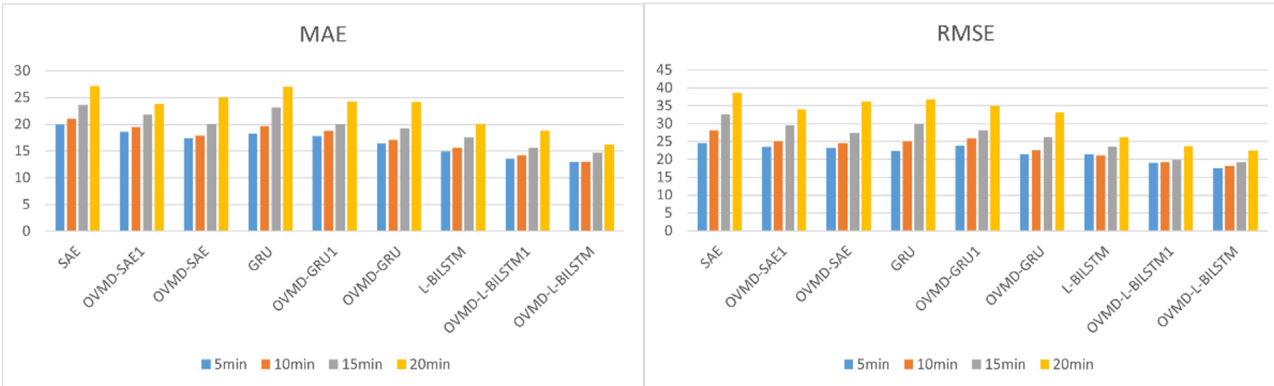

Fig. 19 Comparison of MAE and RMSE values for four time steps.

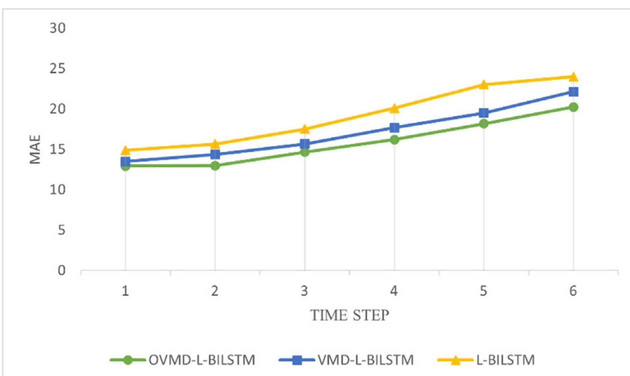

Fig. 20 MAE of OVMD-L-BILSTM and its variant models at 6 time steps.

does not consider whether the performance of the model meets the standard under small sample data such as weekends. We only solved the number of components K and penalty factor C of VMD, and did not add other parameters to the optimization algorithm.

According to the limitations and challenges of this study, the following suggestions might be interesting directions for future work.

(1) Establish a prediction model that comprehensively considers the impact of multiple variables.

(2) Build new prediction methods for small sample conditions, such as adding data at the technical level, and so on.

(3) Establish a new VMD optimization algorithm, consider the new fitness function, and set more VMD parameters as independent variables to solve.

CRediT authorship contribution statement

Guowen Dai: Methodology. **Jinjun Tang:** Conceptualization, Writing – review & editing. **Wang Luo:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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